

Chapter 5: Mathematics Logic

Class 20 – Number Patterns, Sequences & Mathematical Reasoning

Learning Objectives

After completing this lesson, students will be able to:

- Identify different types of number patterns and sequences.
 - Complete missing numbers in a sequence.
 - Recognize even and odd number patterns.
 - Perform skip counting accurately.
 - Arrange numbers in ascending and descending order.
 - Develop logical thinking and mathematical reasoning skills.
 - Apply number patterns in real-life situations.
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Introduction

Mathematics is all about identifying patterns and solving problems logically. Number patterns help us understand how numbers are connected and make calculations easier. Learning sequences, skip counting, and mathematical reasoning improves observation, thinking, and problem-solving abilities.

In this lesson, students will explore different number patterns, solve sequence puzzles, and strengthen their logical thinking through fun mathematical activities.

What are Number Patterns?

A **number pattern** is a sequence of numbers that follows a specific rule.

Examples

- 2, 4, 6, 8, 10...
- 5, 10, 15, 20...
- 100, 90, 80, 70...

Students learn to identify the rule used in each pattern.

Number Sequences

A **number sequence** is a list of numbers arranged according to a rule.

Increasing Sequence

- 10, 20, 30, 40, 50...

Decreasing Sequence

- 100, 90, 80, 70, 60...

Missing Number Sequence

- 5, 10, __, 20, 25
 - Answer: **15**
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Even and Odd Number Patterns

Even Numbers

Even numbers are divisible by **2**.

Examples:

- 2
- 4
- 6
- 8
- 10
- 12

Odd Numbers

Odd numbers are not divisible by **2**.

Examples:

- 1
- 3
- 5
- 7
- 9
- 11

Skip Counting

Skip counting means counting by adding the same number repeatedly.

Skip Counting by 2

2, 4, 6, 8, 10, 12...

Skip Counting by 5

5, 10, 15, 20, 25...

Skip Counting by 10

10, 20, 30, 40, 50...

Skip Counting by 25

25, 50, 75, 100...

Skip Counting by 50

50, 100, 150, 200...

Skip Counting by 100

100, 200, 300, 400...

Ascending and Descending Order

Ascending Order

Numbers arranged from **smallest to largest**.

Example:

12, 18, 25, 39, 47

Descending Order

Numbers arranged from **largest to smallest**.

Example:

90, 75, 60, 45, 30

Missing Numbers

Find the missing numbers by observing the pattern.

Examples

1. 12, 14, __, 18, 20
 2. 30, __, 50, 60
 3. __, 15, 20, 25
 4. 100, 90, __, 70
 5. 5, 10, 15, __, 25
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Shape Patterns

Patterns can also be formed using shapes.

Examples





Students identify the next shape in the pattern.

Logical Number Series

Find the rule and complete the series.

Examples

- 3, 6, 9, 12, __
 - 20, 18, 16, 14, __
 - 7, 14, 21, 28, __
 - 100, 80, 60, 40, __
 - 9, 18, 27, 36, __
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Mathematical Reasoning

Mathematical reasoning helps students think logically before answering.

Example Questions

- Which number comes before 75?
 - Which number comes after 149?
 - Which number is greater: 58 or 85?
 - Which number is smaller: 120 or 102?
 - What is the next number in the pattern?
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Problem-Solving Strategies

Students learn to:

- Observe carefully.
- Find the rule.
- Think step by step.

- Check the answer.
 - Explain their reasoning.
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Number Patterns in Daily Life

Students identify patterns used in daily life.

Examples

- Calendar dates
 - House numbers
 - Bus numbers
 - Classroom roll numbers
 - Numbered parking spaces
 - Clock numbers
 - Sports jersey numbers
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Did You Know?

- Number patterns help computers solve problems quickly.
 - Skip counting makes multiplication easier.
 - Patterns are found in nature, music, art, and mathematics.
 - Logical thinking improves memory and decision-making skills.
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Points to Remember

- A number pattern follows a rule.
- Skip counting helps with quick calculations.
- Even numbers are divisible by 2.
- Odd numbers are not divisible by 2.
- Always identify the pattern before completing a sequence.
- Logical reasoning helps solve mathematical problems correctly.